4.7.1 The graph of the secant function

The graph of secant function in $0 \le x \le 2\pi$, $x \ne \frac{\pi}{2}, \frac{3\pi}{2}$, is shown in Fig. 4.23. In the first and fourth quadrants or in the interval $-\frac{\pi}{2} < x < \frac{\pi}{2}$, $y = \sec x$ takes only positive values, whereas it takes only negative values in the second and third quadrants or in the interval $\frac{\pi}{2} < x < \frac{3\pi}{2}$.

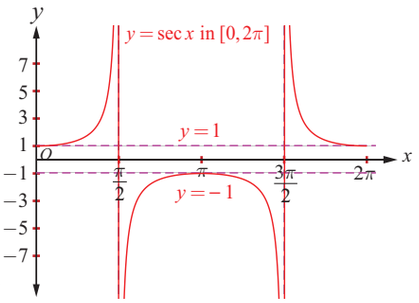


Fig. 4.23

For $0 \le x \le 2\pi$, $x \ne \frac{\pi}{2}, \frac{3\pi}{2}$, the secant function is continuous. The value of secant function raises from 1 to ∞ for $x \in \left[0, \frac{\pi}{2}\right)$; it raises from $-\infty$ to -1 for $x \in \left(\frac{\pi}{2}, \pi\right]$. It falls from -1 to $-\infty$ for $x \in \left[\pi, \frac{3\pi}{2}\right)$ and falls from ∞ to 1 for $x \in \left(\frac{3\pi}{2}, 2\pi\right]$.

As $y = \sec x$ is periodic with period 2π, the same segment of the graph for $0 \le x \le 2\pi$, $x \ne \frac{\pi}{2}, \frac{3\pi}{2}$, is repeated in $[2\pi, 4\pi] \setminus \left\{\frac{5\pi}{2}, \frac{7\pi}{2}\right\}$, $[4\pi, 6\pi] \setminus \left\{\frac{9\pi}{2}, \frac{11\pi}{2}\right\}$, $\cdots$ and in $\cdots$, $[-4\pi, -2\pi] \setminus \left\{-\frac{7\pi}{2}, -\frac{5\pi}{2}\right\}$, $[-2\pi, 0] \setminus \left\{-\frac{3\pi}{2}, -\frac{\pi}{2}\right\}$.

Now, the entire graph of $y = \sec x$ is shown in Fig. 4.24.

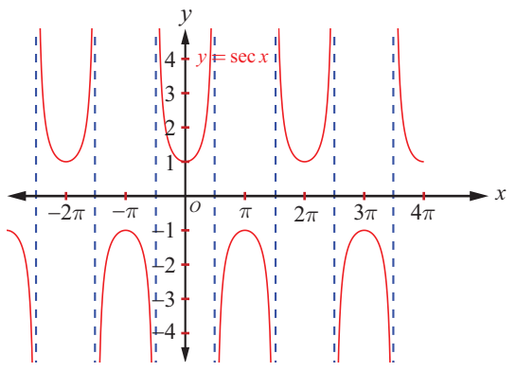


Fig. 4.24

4.7.2 Inverse secant function

The secant function, $\sec x : [0, \pi] \setminus \left\{\frac{\pi}{2}\right\} \to \mathbb{R} \setminus (-1, 1)$ is bijective in the restricted domain $[0, \pi] \setminus \left\{\frac{\pi}{2}\right\}$. So, the inverse secant function is defined with $\mathbb{R} \setminus (-1, 1)$ as its domain and with $[0, \pi] \setminus \left\{\frac{\pi}{2}\right\}$ as its range.

Definition 4.7

The inverse secant function $\sec^{-1} : \mathbb{R} \setminus (-1, 1) \to [0, \pi] \setminus \left\{\frac{\pi}{2}\right\}$ is defined by $\sec^{-1}(x) = y$ whenever $\sec y = x$ and $y \in [0, \pi] \setminus \left\{\frac{\pi}{2}\right\}$.

12th_Maths_EM_Vol1_CH 4 Inverse Trigonometry.indd 150 16-12-2022 13:14:46